

JKBOSE

2026 CLASS 11TH

VECTOR

LECTURE-4

PHYSICS

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Topics To Be Covered

- 1** Revision ✓
- 2** Dot Product ✓
- 3** Resolution of Vector ✓
- 4** ..

“Har Mushkilon se takrayenge,
Jo hamara hain use payenge
Jahan kuch log meri haar chahate hain,
Dekhna ek din jeet kar hum dekhayenge”

Recap from 1st Lecture

Vector

- The pd have magnitude and direction



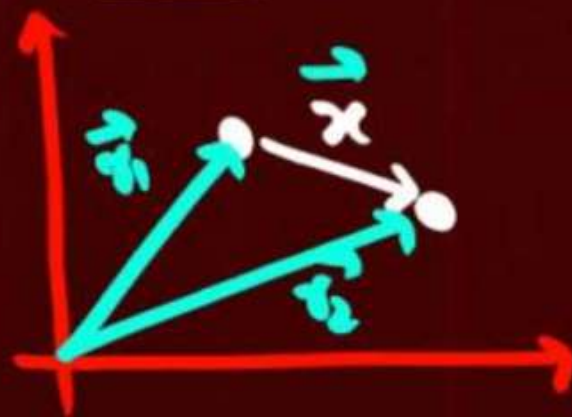
A
→
A

Types of Vector

(i) Position vector



(ii) Displacement V



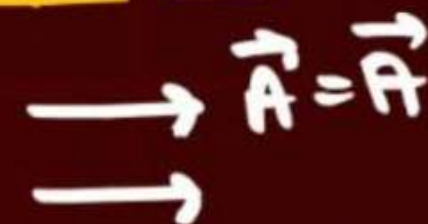
(iii) Parallel vector



(iii) Antiparallel vector



(iv) Equal vector

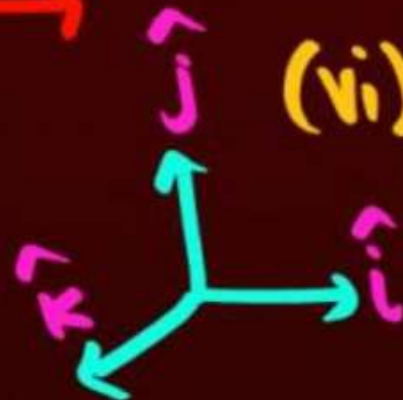


(v) Orthogonal vector



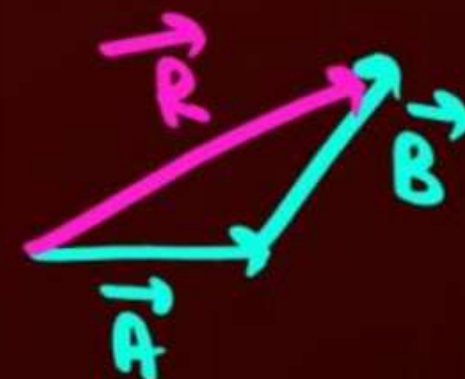
(vi) Unit vector

$n=1$
Direction



+ - × ÷
A B
a b
r a
vector

① Triangular law of vector Addition

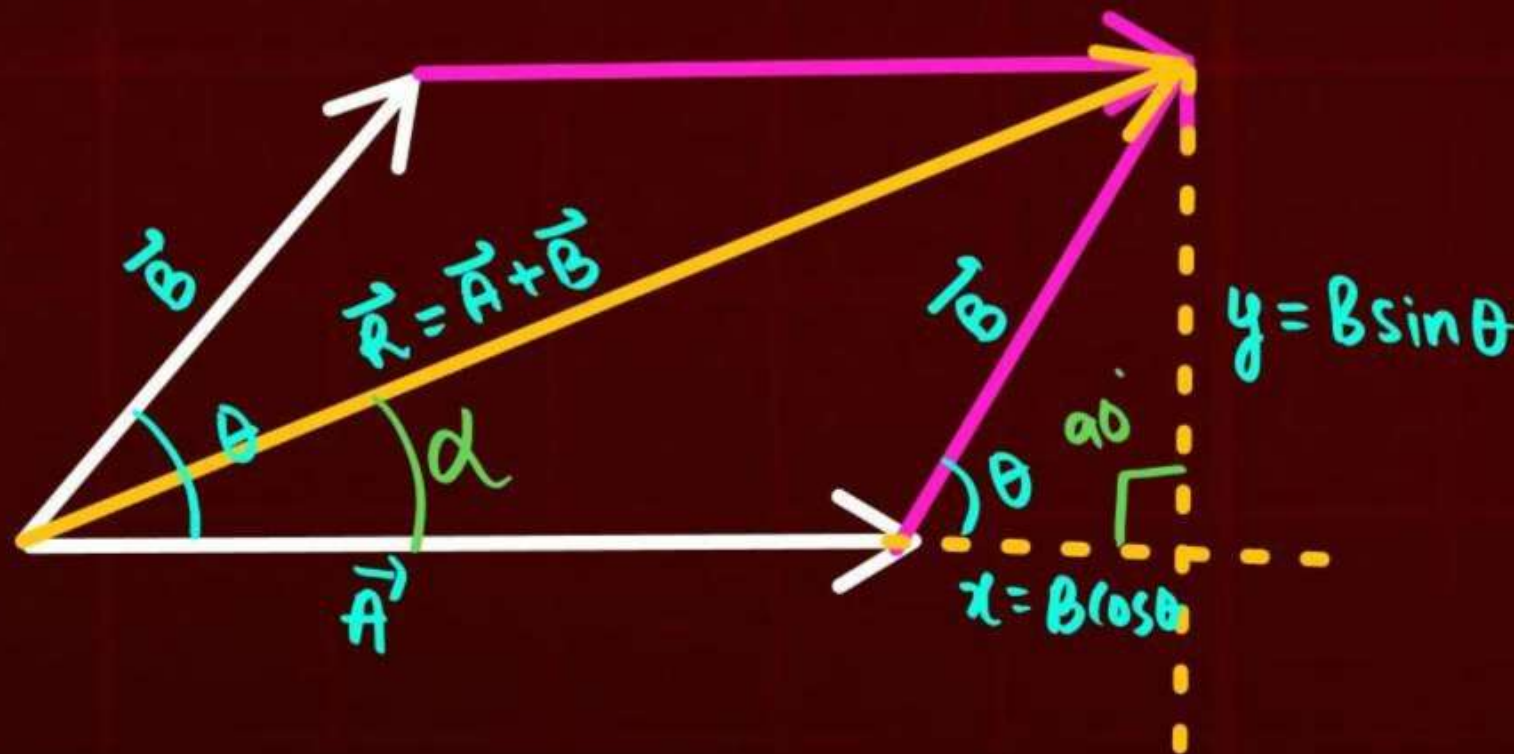


Recap from 2nd Lecture

Parallelogram law of vector Addition

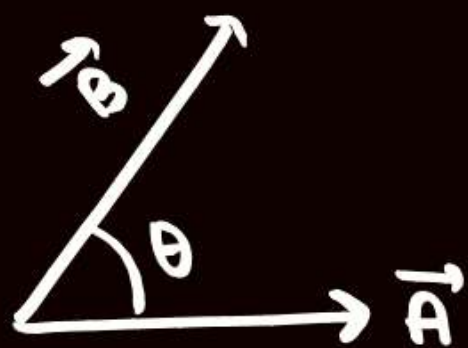
$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

$$\tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}$$

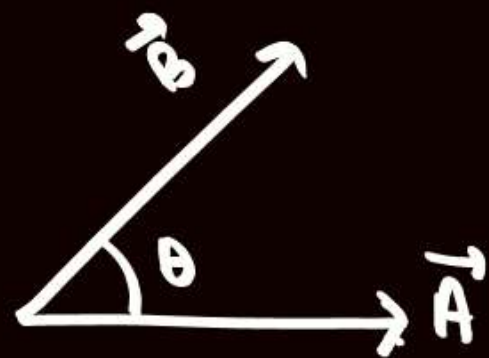


Recap - ③

Cross Product.



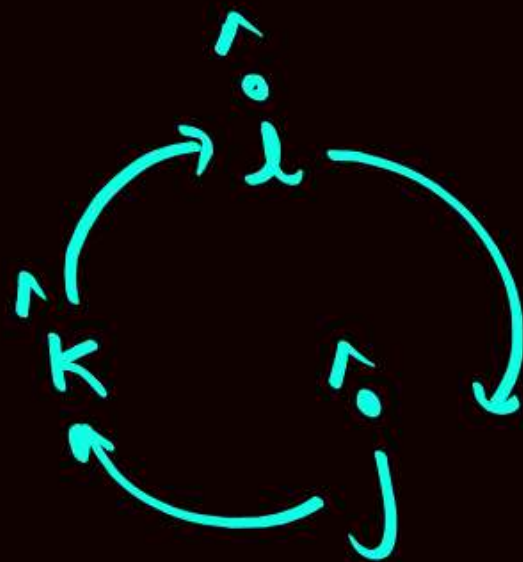
$$\vec{B} \times \vec{A} = \overbrace{BA \sin \theta}^m \hat{n}_i$$



$$\vec{A} \times \vec{B} = AB \sin \theta \hat{n}_o$$

Properties of Cross Product.

- ① It is a vector product.
- ② Direction is given by **Screw Rule**
- ③ $AB \sin 90^\circ = AB \times 1 = \underline{AB}$ (max)
- ④ $AB \sin 0 = AB \times 0 = 0$ (min)
- ⑤ $\vec{A} \times \vec{B} \neq \vec{B} \times \vec{A}$ do not show commutative property



⑥ Distributive property

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$$

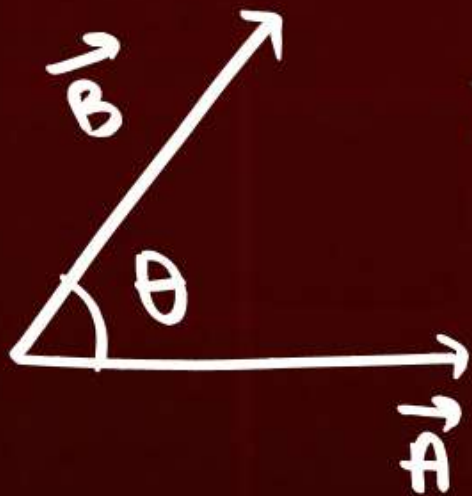
⑥ Dot Product

Scalar product

WORK | Energy | Power



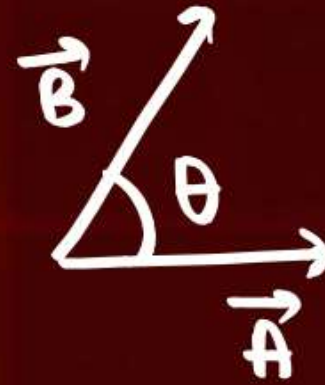
The dot product of two vector is the multiplication of magnitude of two vector and Cos of Angle b/w them.



$$\vec{A} \cdot \vec{B} = \frac{AB \cos \theta}{(m)}$$

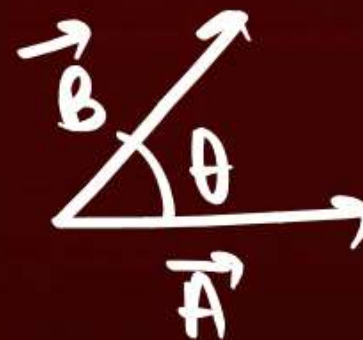
Scalar product.

Case ①



$$\vec{B} \cdot \vec{A} = BA \cos \theta$$

Case ②



$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$\# \vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A} \checkmark$$

no need to find out the Direction

Properties of dot product of two vector

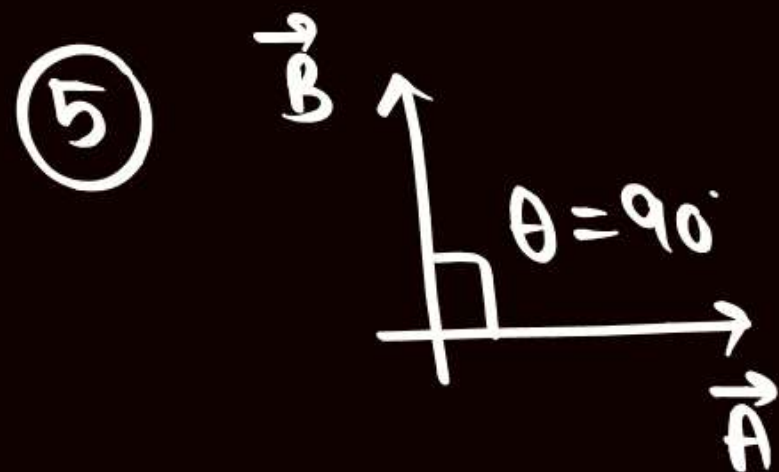
① $\vec{A} \cdot \vec{B} = \underline{AB \cos \theta}$ (Answer $\rightarrow$ Scalar) It is a scalar product.

② no need to find out the Direction.

③ $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$ Shows Commutative property.

④ Distributive property

$$\vec{A} \cdot (\vec{B} + \vec{C}) = \underline{\vec{A} \cdot \vec{B}} + \underline{\vec{A} \cdot \vec{C}}$$



$$\begin{aligned}\vec{A} \cdot \vec{B} &= AB \cos \theta \\ &= AB \cos 90^\circ \\ &= AB \times 0 = \underline{\underline{0}}\end{aligned}$$

⑥



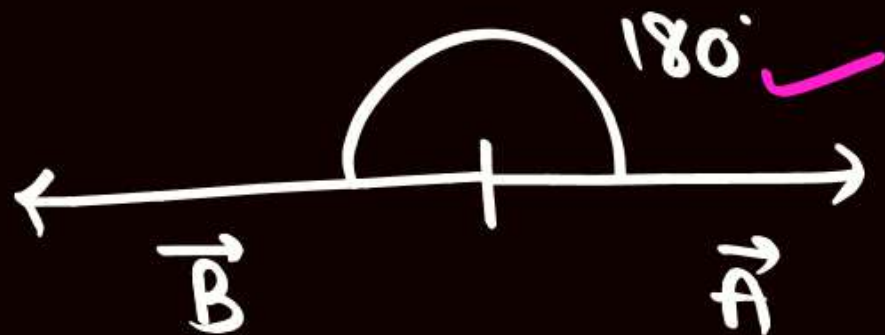
$$\theta = 0^\circ \checkmark$$

$$\vec{A} \cdot \vec{B} = AB \cos 0$$

$$= AB \times 1 = \underline{\underline{+AB}}$$

→ Neet
(max)

⑦



$$\vec{A} \cdot \vec{B} = AB \cos 180^\circ$$

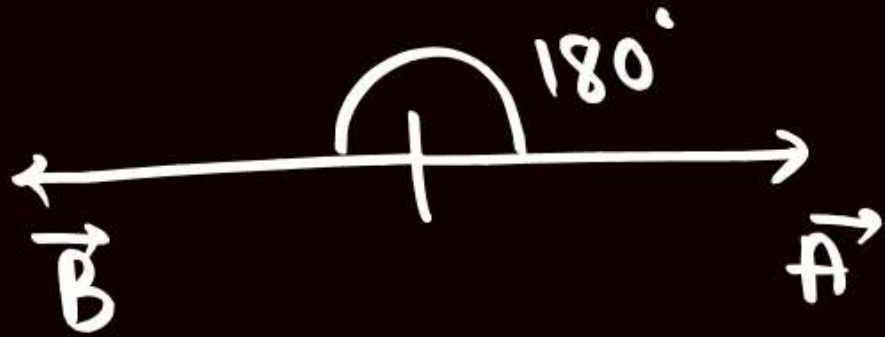
$$= AB \times -1$$

$$= \underline{\underline{-AB}} \checkmark \text{ (mini)}$$

→ Neet
12th

$$-1 < 0$$

Q Numerical (Cross-product)



$$\vec{A} \times \vec{B} = ?$$

Sol

$$\begin{aligned}\vec{A} \times \vec{B} &= AB \sin \theta \\ &= AB \sin 180^\circ \\ &= AB \times 0 \\ &= 0\end{aligned}$$

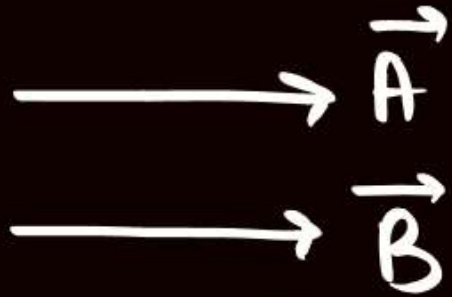
#

$$\theta \rightarrow 0^\circ$$

$$\theta \rightarrow 180^\circ$$

$$\vec{A} \times \vec{B} = \underline{\underline{(\text{minimum})}}$$

Q Numerical



$$\theta = 0$$

$$\vec{A} \times \vec{B}$$

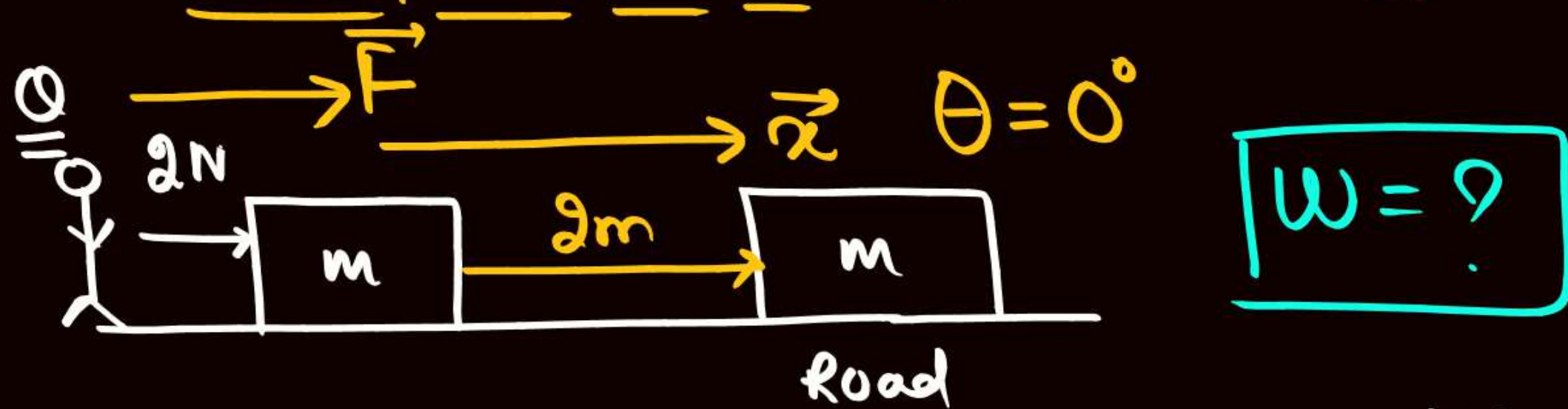
$$\underline{AB} \sin \theta$$

$$AB \sin 0$$

$$\underline{AB} \times \underline{0}$$

$$\underline{\underline{0}} \text{ (minimum)}$$

Feel of Dot product. (Work | Energy | Power)



Sol Work dot product $\vec{F}$ and $\vec{x}$

$$W = \vec{F} \cdot \vec{x}$$

$$W = Fx \cos \theta$$

$$W = Fx \cos 0$$

$$W = Fx \times 1$$

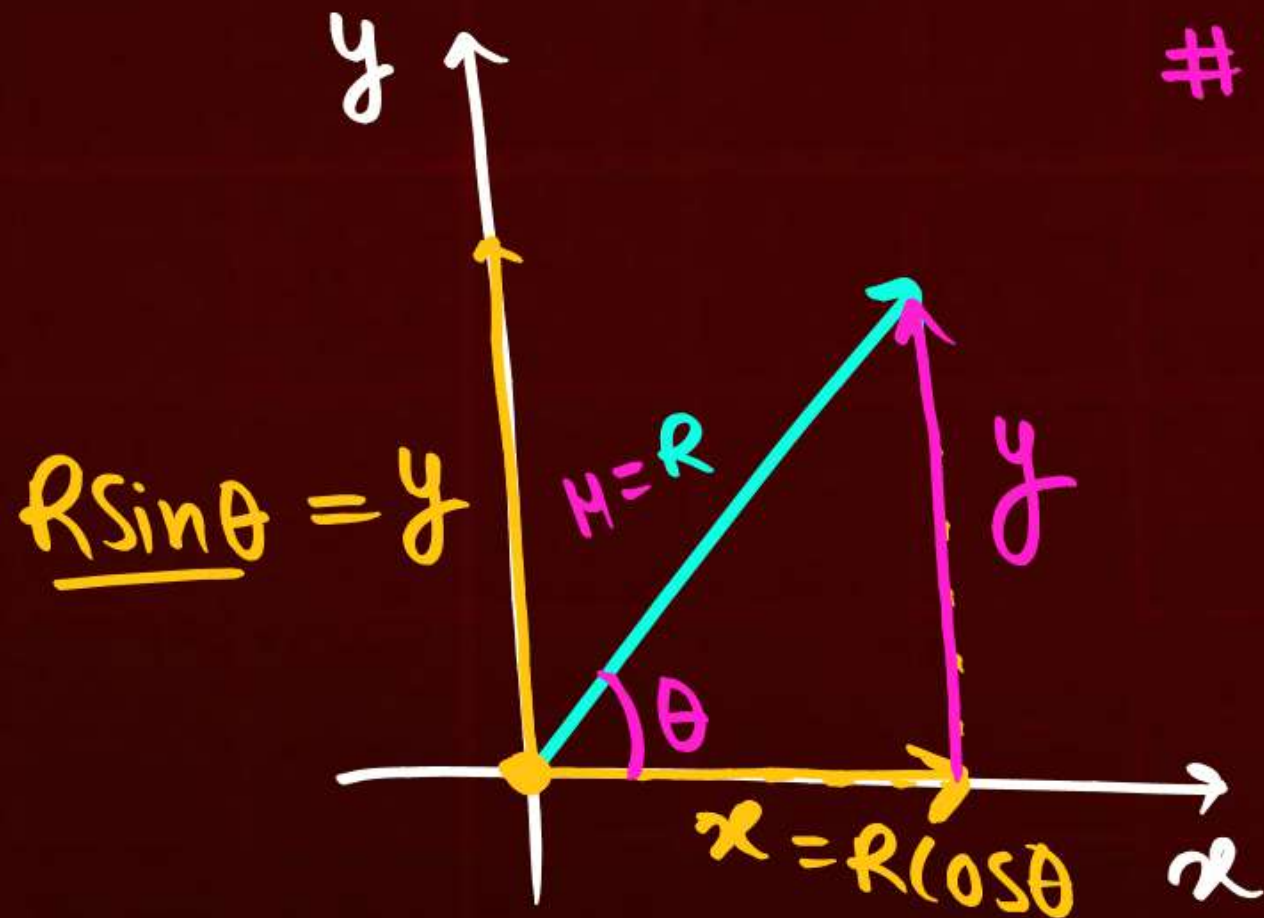
$$\begin{aligned} W &= Fx \\ &= 2 \times 2 \\ &= 4J \end{aligned}$$

Resolution of Vector

2025 | 2024 | 2022 | 2018 | 2017



- # Resultant Vector ko tora into its Component.
- # Break down of Resultant vector into its x and y -component.
- # Rectangular component of vector.



$$\# \sin \theta = \frac{O}{H}$$

$$\sin \theta = \frac{y}{R}$$

$$R \sin \theta = y$$

$$y = R \sin \theta$$

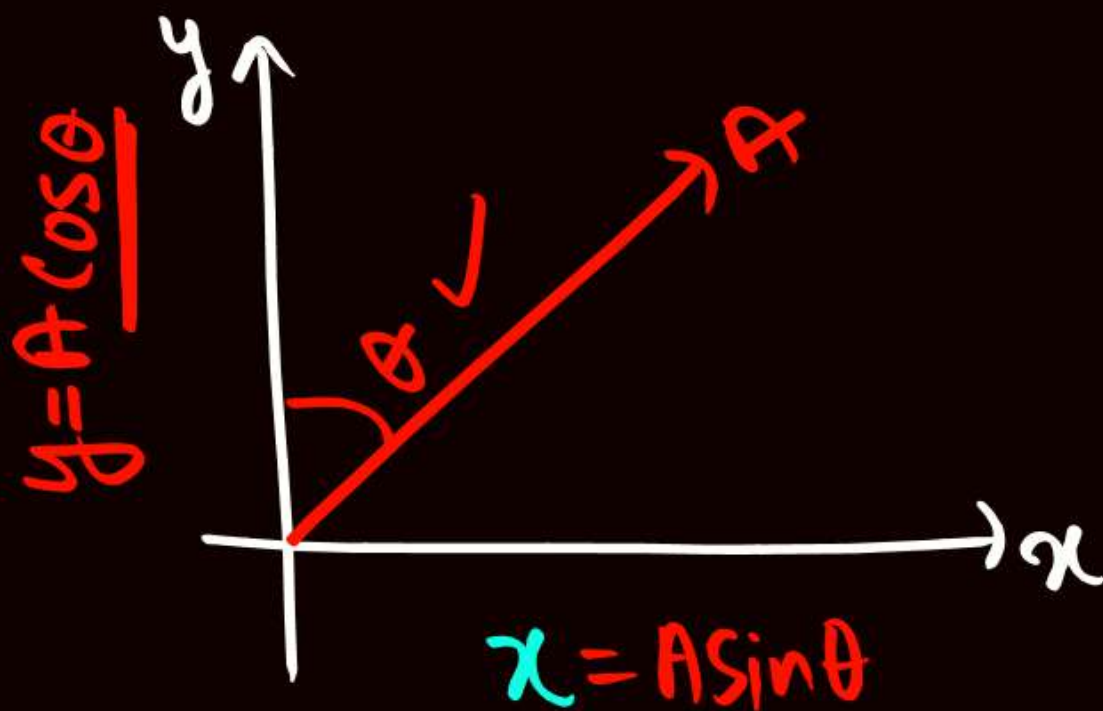
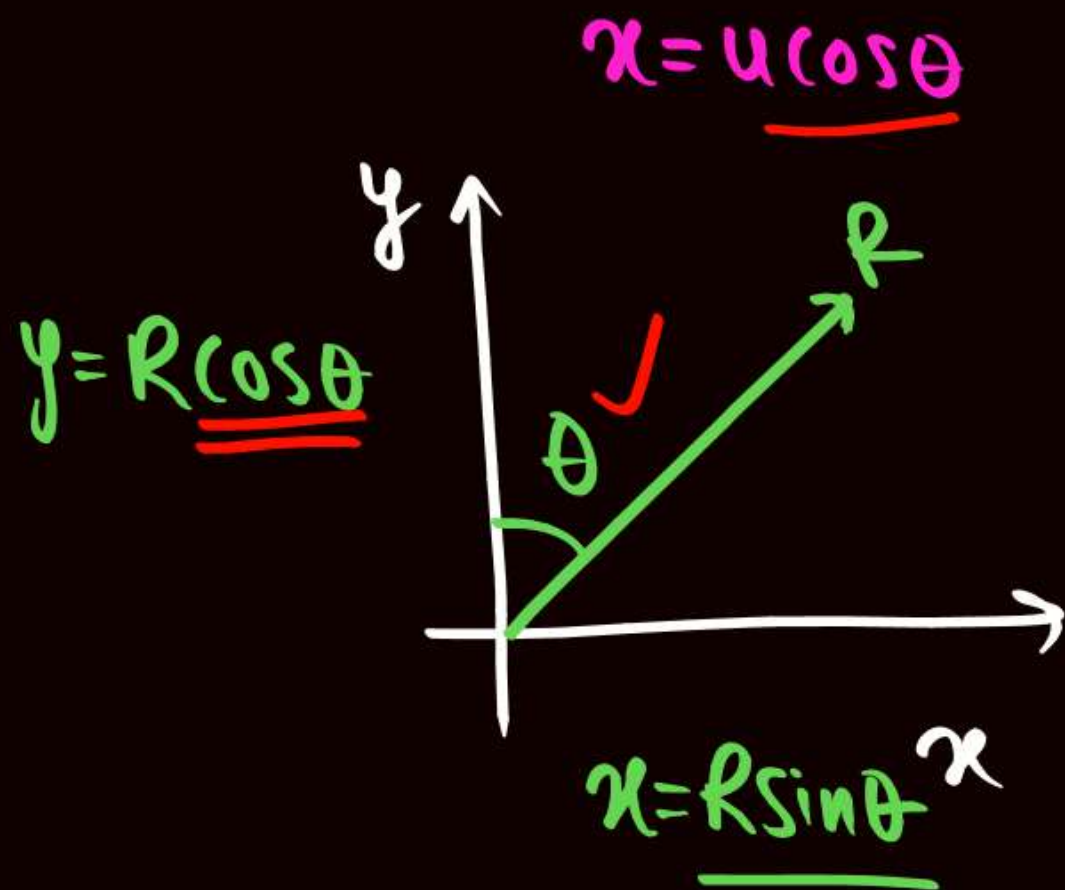
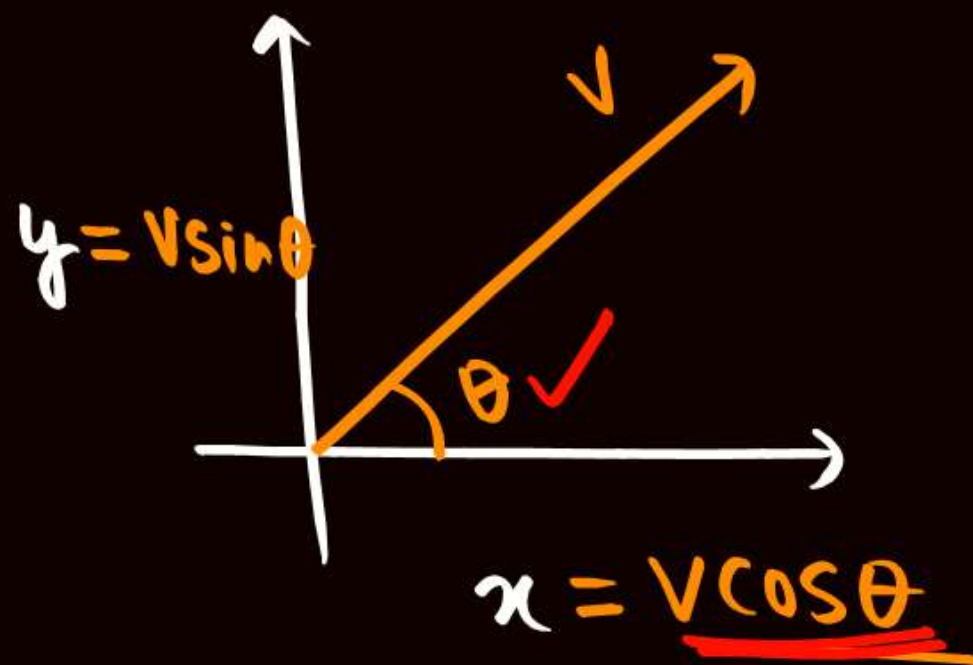
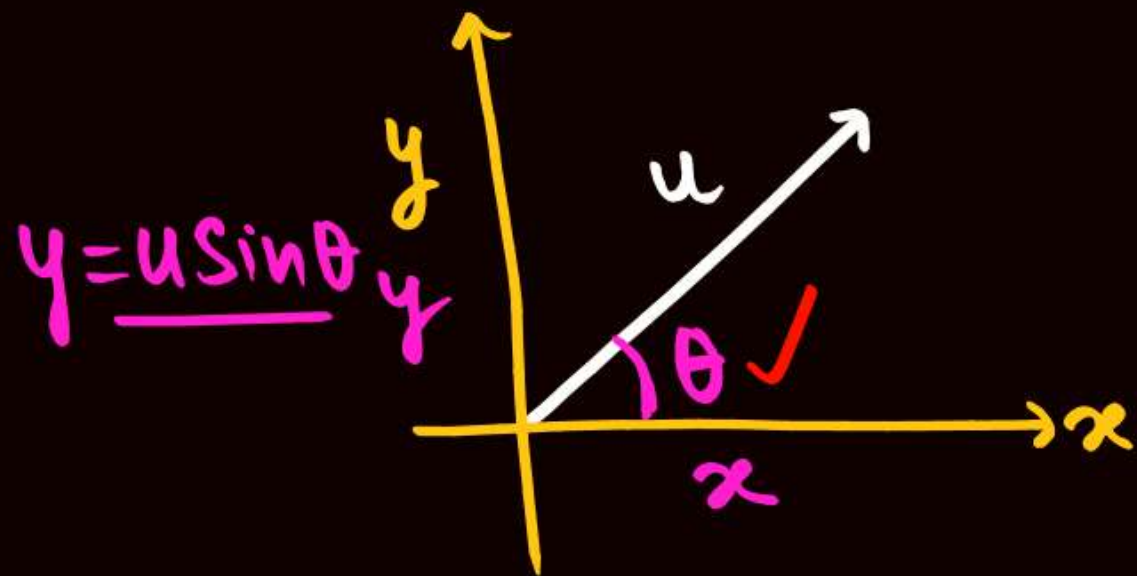
$$\# \cos \theta = \frac{A}{H}$$

$$\cos \theta = \frac{x}{R}$$

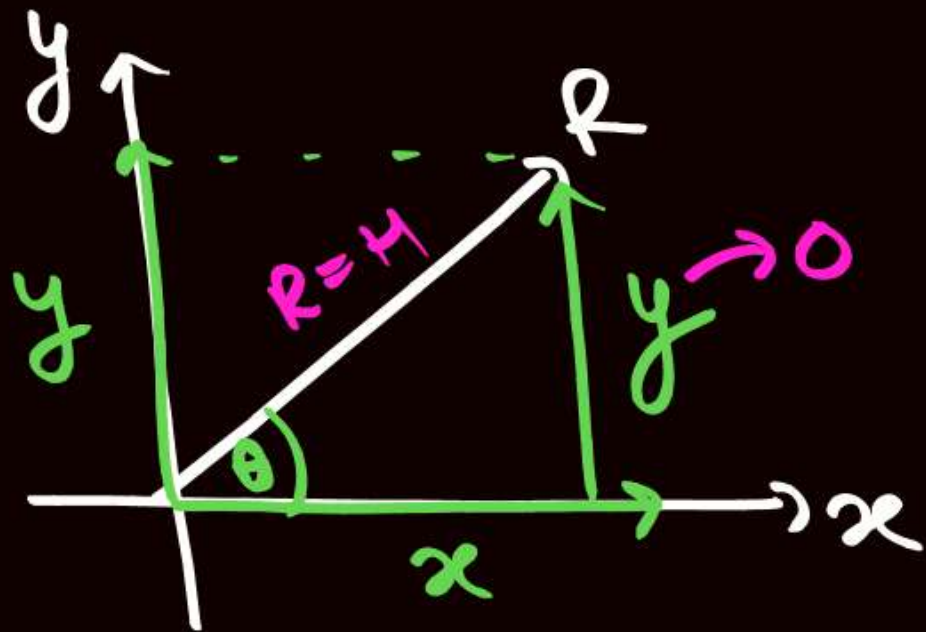
$$R \cos \theta = x$$

$$x = R \cos \theta$$

Q numericals



Resolution of vector



$$\# \sin \theta = \frac{O}{H}$$

$$\sin \theta = \frac{y}{R}$$

$$R \sin \theta = y$$

$$\# \cos \theta = \frac{A}{H}$$

$$\cos \theta = \frac{x}{R}$$

$$R \cos \theta = x$$

Q Neet level $\rightarrow$ 1D Motion

the displacement as the function of time is given by

$$\boxed{x = t^3 + 3}$$

$\downarrow$ variable

Sol

$$v = \frac{dx}{dt}$$

$$v = \frac{d(t^3 + 3)}{dt}$$

$$v = \frac{dt^3}{dt} + \frac{d3}{dt} \hat{=} 0$$

$$v = ?$$

$$v = \frac{dt^3}{dt}$$

$$v = 3t^{3-1}$$

$$v = 3t^2$$

$\downarrow$ variable

$$a = ?$$

$$a = \frac{dv}{dt}$$

$$a = \frac{d}{dt} 3t^2$$

$$a = 2 \times 3t^{2-1}$$

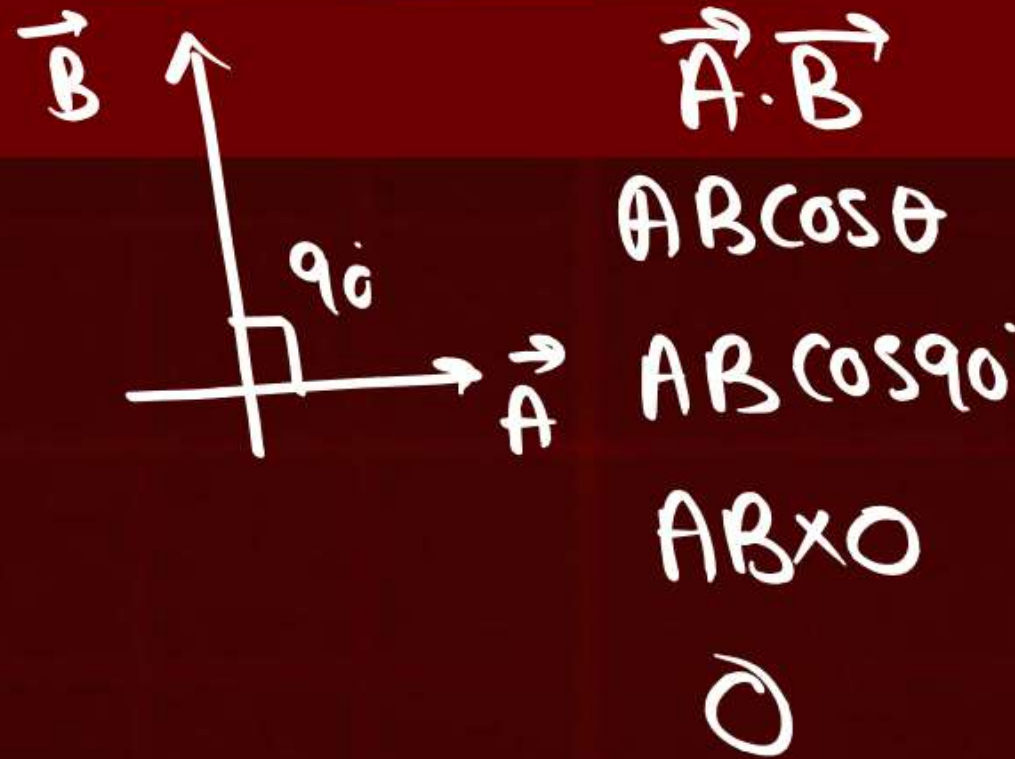
$$a = 6t^1 = \underline{\underline{6t}}$$

Variable

$$\frac{dx^n}{dx} = nx^{n-1}$$

Chocolate Question

The dot product of two perpendicular vectors is:

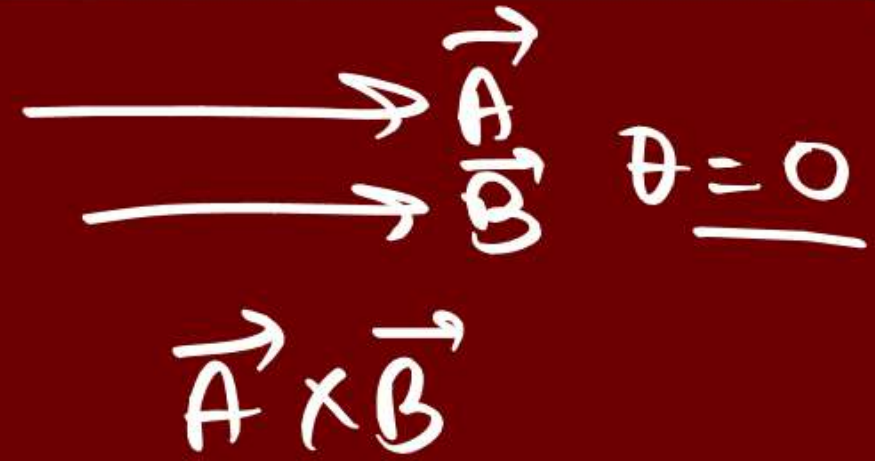


- A** 1
- B** Maximum
- C** 0 ✓
- D** Infinite

Chocolate Question

The cross product of two parallel vectors is:

X



A Maximum ✓

B 0

C 1

D Undefined

$$AB \sin \theta$$

$$AB \sin 0$$

$$AB \times 1$$

$$AB$$

Chocolate Question

The dot product of two vectors gives a:

$$\vec{A} \cdot \vec{B} = \underline{AB \cos \theta}$$

A Vector

B Matrix

C Scalar ✓

D Tensor

Chocolate Question

The cross product of two vectors gives a:

$$\vec{A} \times \vec{B} = AB \sin \theta \hat{n}$$

- A** Scalar
- B** Number
- C** Angle
- D** Vector ✓

Chocolate Question

Which formula represents the dot product?

A ~~$\vec{A} \cdot \vec{B} = AB \cos \theta$~~

B ~~$\vec{A} \times \vec{B} = AB \cos \theta$~~

C ~~$\vec{A} \cdot \vec{B} = AB \sin \theta$~~

D ~~$A \times B = AB$~~

Imp. Information

→ 10min

1D - Revision

Vector → Revision → 10min

Handwritten notes

SUMMARY

Dot product

Resolution of vector

HOMework



Copy check Rs-50/-

$$\vec{\omega} \times \vec{H} = \underline{\hspace{2cm}}$$

$$\vec{\omega} \cdot \vec{H} = \underline{\hspace{2cm}}$$

Thank
You